# Predicting Metabolic Cost During Human-in-the-Loop Optimization

**Course:** UE24CS352A – Machine Learning
**Team Members1:** Parinitha Acharya-PES2UG24CS339

**Team Members2-**Ritesh-PES2UG24CS340
**Institution:** PES University

## 1. Problem Statement

Human-in-the-loop optimization can be used to find suitable control parameters for wearable robotic devices such as ankle exoskeletons. A key measure of exoskeleton effectiveness is the **metabolic cost of walking**. Directly measuring metabolic cost requires specialized equipment and can be time-consuming.

The objective of this project is to investigate whether machine learning can predict metabolic cost from physiological and exoskeleton-related signals collected during walking. Since metabolic cost is a continuous numerical value, this is formulated as a **regression problem**.

## 2. Dataset

The project uses a publicly available dataset associated with the study *"Metabolic cost of walking with electromechanical ankle exoskeletons under proportional myoelectric control on a treadmill and outdoors."*

The dataset contains metabolic, EMG, and exoskeleton measurements from human walking trials. The available conditions include powered exoskeleton walking, unpowered exoskeleton walking, boots-only walking, standing, and outdoor walking.

For this project, three indoor treadmill conditions were used:

- **Exopowin:** Powered exoskeleton
- **Exounpowin:** Unpowered exoskeleton
- **Noexoin:** Boots only

The final machine-learning dataset contained **21 matched trials from 12 subjects**.

## 3. Data Processing and Feature Extraction

For each subject, the standing trial was used to obtain a baseline oxygen-consumption value. For each indoor walking trial, the final **3 minutes** were selected as the steady-state period.

The target was calculated as:

**Net Metabolic Cost = Walking VO2/Kg – Standing VO2/Kg**

The EMG and exoskeleton data were processed for the corresponding matched trials. Eight signals were used:

- Soleus EMG
- Tibialis Anterior EMG
- Lateral Gastrocnemius
- Medial Gastrocnemius
- Ankle Angle
- Motor Current
- Motor Voltage
- Motor Angle

For each signal, five statistical features were calculated:

**Mean, Standard Deviation, RMS, Minimum, and Maximum.**

Therefore, the final input consisted of:

**8 signals × 5 features = 40 features**

## 4. Machine Learning Methodology

Two regression models were implemented:

### Lasso Regression

Lasso was used as a linear baseline model with regularization to reduce the influence of less important features.

### Random Forest Regression

Random Forest was selected because it can capture nonlinear relationships between physiological signals and metabolic cost.

Since multiple trials may belong to the same subject, random splitting could cause data leakage. Therefore, **subject-based GroupKFold cross-validation** was used so that the model was evaluated on subjects not used during training.

The models were evaluated using:

- **MAE (Mean Absolute Error)**
- **RMSE (Root Mean Squared Error)**
- **$R^2$ Score**

**5. Results**

| **Model** | **MAE** | **RMSE** | **$R^2$** |
|---|---|---|---|
| Lasso Regression | 4.3847 | 4.8241 | -2.8449 |
| **Random Forest** | **3.1039** | **3.2904** | **-0.7888** |

Random Forest performed better than Lasso Regression, giving lower MAE and RMSE and a better $R^2$ value.

However, the Random Forest $R^2$ was still negative. This shows that prediction for completely unseen subjects remains difficult. The main reasons are the small dataset size, only 21 matched trials, and substantial physiological differences between subjects.

**6. Implementation**

The project was implemented in **Python using Visual Studio Code and Jupyter Notebook**.

The main libraries used were:

- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- OpenPyXL
- Joblib

The workflow consists of:

**Data Collection → Data Preprocessing → Baseline Calculation → EMG Feature Extraction → Dataset Creation → Model Training → Subject-Based Validation → Performance Evaluation**

The final Random Forest model was saved using Joblib. Visualizations such as actual-versus-predicted plots and feature-importance plots were also generated.

The source code is maintained in a **private GitHub repository** with a README containing project setup and execution instructions, as required by the assignment. e353b7a7-f926-4caf-ad61-9c78ed7...

## 7. Conclusion

This project demonstrates a machine-learning approach for predicting metabolic cost from EMG and exoskeleton-related signals during human walking.

A dataset containing **21 matched trials from 12 subjects** was processed to obtain 40 signal-based features. Lasso Regression and Random Forest Regression were evaluated using subject-based cross-validation. Random Forest performed better, achieving an MAE of **3.1039** and RMSE of **3.2904**.

Although Random Forest was the better model, its negative $R^2$ indicates that the current model does not generalize strongly to unseen subjects. This highlights the difficulty of predicting human physiological responses using a small dataset.

Future improvements could include using more subjects and trials, additional EMG and gait features, frequency-domain features, better feature selection, and additional machine-learning models.

## References


1. Hybart, R., Villancio-Wolter, K. S., & Ferris, D. P. *Metabolic cost of walking with electromechanical ankle exoskeletons under proportional myoelectric control on a treadmill and outdoors*.
2. Krimsky, E. & Ng, E. *Predicting Metabolic Cost During Human-in-the-Loop Optimization*.
3. Figshare EMG/Exoskeleton Dataset: https://doi.org/10.6084/m9.figshare.22572868.v3
4. Figshare Metabolic Dataset: https://doi.org/10.6084/m9.figshare.22572841.v4